Week 11 Monday

Dictionaries: A Dictionary of Dictionaries

CS1

Essential Question

How do we find one thing by name, and can a dictionary hold other dictionaries?

Week 11 plan; `lessons/07-collections.md`; AI Fluency Lens 11 (Verify) tie-in

Rock-Paper-Scissors: Rules as Data

```
rules = {
 "rock": {"rock": "tie", "paper": "lose",
          "scissors": "win"},
 "paper": {"rock": "win", "paper": "tie",
           "scissors": "lose"},
 "scissors": {"rock": "lose", "paper": "win",
              "scissors": "tie"},
}
print(rules["rock"]["scissors"])  # win
```

- A dictionary whose values are dictionaries: `rules[player][opponent]`.
- First key picks the row, second key picks the column. Two lookups, one answer.
- Predict first: what is `rules["paper"]["rock"]`?

Walk It, Score It, Guard It

```
for mine in rules:                  # outer keys
    for theirs in rules[mine]:  # inner keys
        print(mine, theirs, rules[mine][theirs])

score = {"win": 0, "lose": 0, "tie": 0}
score[judge("rock", "scissors")] += 1
```

- A second dictionary keeps score: keys are results, values are counts.
- A move that is not a key raises `KeyError`; check with `in` first.
- **Verify (Lens 11):** try an empty string, a capital letter, a move not in the table.
- **Run it, read the walkthrough:** Canvas, Week 11 Overview page, "Rock-Paper-Scissors example"